# Temperature and particle motion

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a particle model to distinguish average motion from total energy.

**SUCCESS CRITERIA - I can...**

1. Describe temperature in terms of microscopic motion.
2. Use kelvin in proportional temperature comparisons.
3. Avoid treating a single particle speed as the temperature.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Sunlight carries energy across 3 million km of nearly empty space in a simplified example. Identify the transfer mechanism and explain why matter is unnecessary.

**Marzano check - circle:** ① ② ③

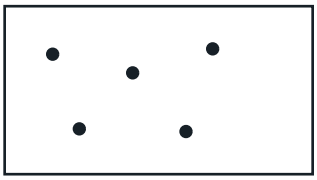
## 02 Worked example: explain the model

**Reference:** $K_{avg} \propto T$; $T(K) = T(°C) + 273.15$.
Ideal gas: mean translational kinetic energy per particle scales with absolute temperature. ∝ means proportional to.

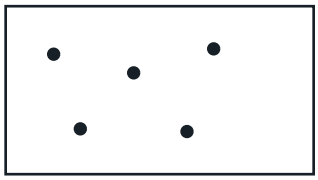
**Worked reasoning:** In a gas model, warmer temperature corresponds to greater average random molecular kinetic energy. Individual particles still have a range of speeds.

**Your turn:** Draw two equal-size samples using dots and arrows; label arrows as particle velocities and make the warmer sample's average arrow length greater.

Sample A



Sample B



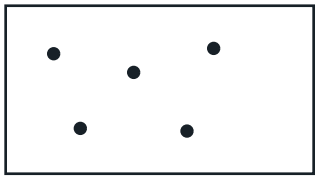
Add velocity arrows; use lengths to show a range of speeds.

**Marzano check - circle:** ① ② ③

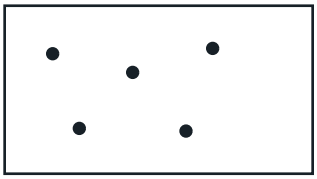
## 03 Read and interpret the model

Each box represents a sample of the same ideal gas. One is cooler and one warmer. Add particle-velocity arrows with a range of lengths in each box. Make the warmer sample's average kinetic energy greater, without making every particle identical.

Sample A



Sample B



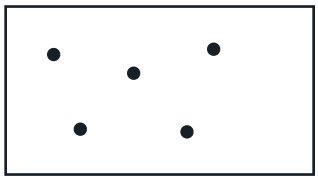
Add velocity arrows; use lengths to show a range of speeds.

**Marzano check - circle:** ① ② ③

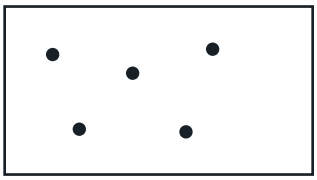
## 04 Apply the physics

An ideal-gas simulation warms a sample from 300 K to 600 K. By what factor does average translational kinetic energy per particle change? Label the two particle models to show the comparison. Do not confuse energy with speed.

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

A student doubles 20°C to 40°C and claims average particle kinetic energy doubles. Explain why the scale matters.

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Two students perform the same lift. One takes 4 times as long as the other. Compare work and average power. Use the table to distinguish energy from its transfer rate.

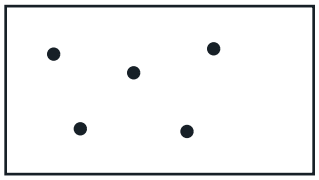
| Case | Energy (J) | Time (s) | Power (W) |
|---|---|---|---|
| A | ____ | ____ | ____ |
| B | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3

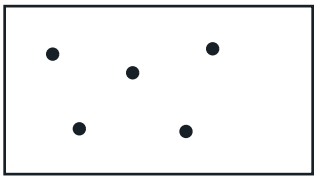
## 07 Exit check

At equal kelvin temperature, does a gas sample with twice as many particles have twice the average energy per particle?

Sample A



Sample B



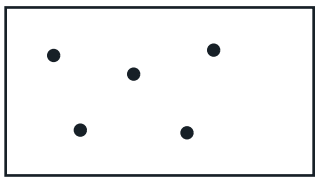
Add velocity arrows; use lengths to show a range of speeds.

**Marzano check - circle:** 1 2 3

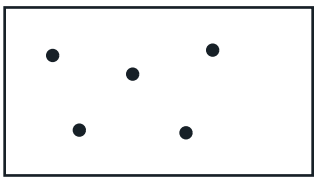
## 08 STRETCH: test the model

Two gas samples have the same temperature but different volumes. Must the particle speeds differ? Explain what temperature alone fixes in this model.

Sample A



Sample B



Add velocity arrows; use lengths to show a range of speeds.

**Marzano check - circle:** 1 2 3